Year II — Algebra

Based on lectures by Various Lecturers

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Chapter 1

Rings, subrings and fields

1.1 Definition of a Ring

Before being able to define a ring, we must first redefine *binary operations*:

Definition 1.1.1 (Binary Operation)**.** A *binary opeation* is a function that takes two elements as an argument and produces a single new output value. Examples include addition, multiplication and subtraction.

Remark 1.1.2. All rings form Abelian groups (review first year if unsure what this means)

Definition 1.1.3 (Ring)**.** A ring R is a set together with two binary operations (usually, but not always, written,$+\cdot$) called "addition" and "multiplication", such that $+$ makes R into an abelian group, and such that the following conditions hold:

i) **Existence of an Identity** - There exists an element $1 \in R$ such that $1 \cdot x = x \cdot 1 = x$, for all $x \in R$.

ii) **Associativity** - For any $x, y, z \in R$ we have $x(yz) = (xy)z$.

iii) **Distributivity** - For any $x, y, z \in R$ we have $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

1.2 Examples of Rings

– Let V be a finite dimensional vector space over $\mathbb{C}$. Let

$$\mathrm{End}(V) = \{f : V \to V \mid f \text{ is a linear map } \}.$$

An element of $\mathrm{End}(V)$ is usually called a (linear) endomorphism. We can add two such functions like we have seen before:

$$(f + g)(v) = f(v) + g(v).$$

Now we can't directly "multiply" two endomorphisms, because V does not have any multiplication (V is not in general a ring). However, we can compose two endomorphisms

$$(f \circ g)(v) = f(g(v)),$$

and composition is a binary operation on V. From Linear Algebra I, we already know that $\mathrm{End}(V)$ is an abelian group under addition. To check that this is a ring, we need to check the three properties:

Identity element is provided by the identity map $Id : V \to V$. Associativity and distributivity are also not hard to check. For example we need to establish that if $f, g, h \in \mathrm{End}(V)$ we have

$$f \circ (g \circ h) = (f \circ g) \circ h.$$

That is, for every vector $v \in V$ we have that $(f \circ (g \circ h))(v) = ((f \circ g) \circ h)(v)$. For the definition of the "multiplication" we have that the left hand side is equal to

$$(f \circ (g \circ h))(v) = f((g \circ h)(v)) = f(g(h(v))$$

and

$$((f \circ g) \circ h)(v) = (f \circ g)(h(v)) = f(g(h(v))$$

That is the two endomorphisms are having the same effect on any vector v of V. That is they are the sane maps. Similarly one checks distributivity.

- $\mathbb{N}$ is not a ring $(1 - 2 \notin \mathbb{N})$
- $\mathbb{Z}$ is a ring but not a field: $\frac{1}{2} \notin \mathbb{Z}$.
- $\mathbb{Q}[x]$ is a ring but not a field: $\frac{1}{x} \notin \mathbb{Q}[x]$.
- $\mathrm{M}_n(\mathbb{Q})$ is a ring but not a field: there is no way we can make sense of division. For example, $\frac{1}{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}$ is meaningless since that matrix does not have an inverse.
- The set $C(\mathbb{R})$ is a ring but not a field: the constant function $1 : \mathbb{R} \to \mathbb{R}$ belongs to the set, and so does the identity function Id defined by $\mathrm{Id}(x) = x$ for all $x \in \mathbb{R}$. However, the function $\frac{1}{\sqrt{d}}$ is not continuous at $x = 0$, so does not belong to $C(\mathbb{R})$.

Method 1.2.1 (Determining a Ring)**.** To determine a ring, one should first check that the ring forms an abelian group with respect to the 'addition' operation and that the following requirements hold for the 'multiplication' operation:

i) **Existence of an Identity** - There exists an element $1 \in R$ such that $1 \cdot x = x \cdot 1 = x$, for all $x \in R$.

ii) **Associativity** - For any $x, y, z \in R$ we have $x(yz) = (xy)z$.

iii) **Distributivity** - For any $x, y, z \in R$ we have $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

1.3 The integers modulo n

Definition 1.3.1 (Equivalence Relation)**.** A (binary) relation $\sim$, on a set S is said to be an equivalence relation, if and only if it is reflexive, symmetric and transitive. That is, for all a, b, and c in the set S we have:

i) $a \sim a$ (reflexivity).

ii) $a \sim b$ if and only if $b \sim a$ (symmetry).

iii) If $a \sim b$ and $b \sim c$ then $a \sim c$ (transitivity).

Definition 1.3.2 (Equivalence Class)**.** The equivalence class of a under the given relation $\sim$, denoted by $[a]$ (or $\bar{a}$) is defined as

$$[a] = \{x \in S : x \sim a\}.$$

Note that for two $a, b \in S$ with $a \sim b$ we have $[a] = [b]$. In particular when we write $[a]$ we call a the representative of the class $[a]$. Clearly an equivalence class may have more than one representative.

From these structures we can construct another set:

$$S/\sim := \{[a] : a \in S\}$$

That is, this is the set of equivalence classes on the set S with respect to the given equivalence relation $\sim$.

Definition 1.3.3 (Integers Modulon n)**.** The integers modulo n. Let $n \in \mathbb{N}$ be a natural number. Let us consider the set of integer numbers $\mathbb{Z}$ and define the following relation: two integers a and b are equivalent (i.e. $a \sim b$ in the above notation) if n divides $a - b$. The most common notation for this is $a \equiv b \pmod{n}$.

Proof of 'the relation is an equivalence relation'.

- it is reflexive as $a - a = 0$ and hence divisible by n,
- it is symmetric as n divides $a - b$ if and only if n divides $b - a = -(a - b)$.
- And finally for the last condition we consider $a, b, c \in \mathbb{Z}$ and further assume that n divides $a - b$ and $b - c$, that is $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$. That is, $a - b = kn$ and $b - c = rn$ for some $k, r \in \mathbb{Z}$. But then $a - c = a - b + b - c = (k + r)n$ and hence n divides $a - c$ i.e. $a \equiv c \pmod{n}$.

□

Definition 1.3.4 (Residue Class)**.** For an integer a we denote by $\bar{a}$ (this is the most commonly used notation) the equivalence class of a, that is

$$\bar{a} = \{x \in \mathbb{Z} : n \text{ divides } x - a\} = \{x \in \mathbb{Z} : x = a \pmod{n}\}$$

Now note that when we divide an integer by n we will always get a remainder among the numbers

$$0, 1, , \ldots, n - 1.$$

That is, every integer belongs to one of the following equivalence classes:

$$\{\overline{0}, \overline{1}, \ldots, \overline{n-1}\},$$

We will denote this set (of equivalence classes) by $\mathbb{Z}/n$. The elements of $\mathbb{Z}/n$ are called residue classes.

Remark 1.3.5. Make sure that if you ever write $\mathbb{Z}_n$ that you explain that you mean $\mathbb{Z}/n$ as it can cause confusion.

1.4 Subrings and Fields

Definition 1.4.1 (Subring)**.** A *subring* S of a ring R is a set $S \subseteq R$ such that

i) $0 \in S, 1 \in S$ where 0 is the 'additive' identity and 1 it the 'multiplicative identity',

ii) $a, b \in S \Rightarrow a + b \in S$,

iii) $a, b \in S \Rightarrow ab \in S$,

iv) $a \in S \Rightarrow -a \in S$.

Remark 1.4.2 (Intuitive definition of Subring)**.** We can shorten this into an informal definition of 'A subring contains 0_R, 1_R and is closed under addition, subtraction and multiplication.

Remark 1.4.3 (Subrings are Rings)**.** A subring of any ring is a ring itself and is equipped with addition and multiplication of the original ring.

Method 1.4.4 (Showing a Subring)**.** We can quickly show a subring by showing:

- the additive and multiplicative identities exist in the set,
- it's closed under addition
- it's closed under multiplication
- it contains additive inverses.

Method 1.4.5 (Determining a Ring quickly)**.** Rather than displaying a set satisfies all of the properties of a ring, it is sufficient to show that the set is a subring of a ring. It is therefore a ring itself.

Definition 1.4.6 (Field)**.** A ring R is called a field if:

i) R is commutative,

ii) $1 \neq 0$ in R (i.e., R has at least two elements),

iii) For any $a \in R$ such that $a \neq 0$, there exists a $b \in R$ such that $ab = 1$ (b is called the inverse of a).